

CS420 Project: Simplified Healthcare Management System

Bjorn Addington

Task 1 – Entity and Attribute Definition:

1.) Clearly list all attributes:

There are 6 total provided entities, they are as follows:

- a. Hospitals
- b. Departments
- c. Doctors
- d. Patients
- e. Appointments
- f. Medications

For each of them I need to list all possible attributes that would be used. I will list them in order of the a-f list above starting at a.

A:Hospital attributes:

- Hospital-Id
- Hospital Name
- Hospital Address

B:Department attributes:

- Department Name(Dependent and weak entity. I went back and changed it since I used to have it as strong but after task 3 being weak makes more sense)
- Location in hospital
- Phone number

C:Doctor Attributes:

- Doctor-Id
- Full Name
- Specialty
- Email

D:Patient attributes:

- Patient-Id

- Full Name
- Emergency contact
- DOB
- Contact Information

E:Appointment attributes:

- Appointment-Id
- Date and Time
- Reason

F:Medication attributes:

- Medication-Id
- Name
- Allergens

- 2.) Underline the primary key: I have identified that each entity needs a unique Id, this Id will be used as the primary key so that each entity is identifiable.
- 3.) Identify composite attributes: There are a few things such as Hospital address could be made up of many different parts like Zip code or street name. Also, I put Full Name a few times, this is a composite attribute as it is made up of both the first name and the last name. Also, the date and time attribute for the appointment attributes. I figure that can be stored as one attribute, but it is two.
- 4.) Identify multivalued attributes: Contact information for the entity patient is multivalued as there can be several instances of different contact information. Like phone number and email or two phone numbers. Outside of the unique primary keys there can be an argument for almost every attribute to be multivalued as there will always be a patient who could have two of some attribute or a medication that has two different durations for a circumstance that makes sense. However, for simplicity I have only identified the contact information attribute to be multivalued.
- 5.) State whether the entity is strong or weak and justify your answer: Every entity here as I have modeled them is a strong entity as they each have their own globally unique primary key that can identify the relationship to find the entity without another entity that acts as a dependency upon it. For example, the department of a hospital will have non-unique names when more than one hospital is taken into account. Thus, a unique key I believe is necessary. However, task 3 gets into this deeper and will be something I cover later. For this part of task 1 they are all strong as I have modeled them as such.

Task 2 - Relationship Identification:

There is a list of 7 provided relationships. I will model each of them according to the 5 requirements above. Starting from the top of the list and working my way down:

Hospital-Department:

- 1.) Provide the relationship name:
 - a. Hospital section
- 2.) List participating entities:
 - a. Hospital
 - b. Department
- 3.) Specify cardinality:
 - a. 1:N
- 4.) Specify participation:
 - a. Total participation for Hospital
 - b. Total participation for Department
- 5.) If applicable, identify relationship attributes and justify why they belong to the relationship:
 - a. There are no relationship attributes for this relationship.

Department-Doctor:

- 1.) Provide the relationship name:
 - a. Department Depending
- 2.) List participating entities:
 - a. Department
 - b. Doctor
- 3.) Specify cardinality:
 - a. 1:N
- 4.) Specify participation:
 - a. Total for doctors as every doctor must have a department
 - b. Partial for department as there could be departments that do not need doctors like HR or cleaning services.
- 5.) If applicable, identify relationship attributes and justify why they belong to the relationship:
 - a. None for this relationship

Department–Head Doctor:

- 1.) Provide the relationship name:
 - a. Head Doctor assignment
- 2.) List participating entities:
 - a. Department
 - b. Head Doctor
- 3.) Specify cardinality:
 - a. 1:1
- 4.) Specify participation:
 - a. Total department participation as every department must have one head doctor
 - b. Partial doctor participation as not every doctor is a head doctor
- 5.) If applicable, identify relationship attributes and justify why they belong to the relationship:
 - a. As I modeled in task one there should be no need for relationship dependencies.

Doctor–Doctor (supervision):

- 1.) Provide the relationship name:
 - a. Supervisorship
- 2.) List participating entities:
 - a. Doctor role as entity
 - b. Doctor role as supervisor
- 3.) Specify cardinality:
 - a. 1:N
- 4.) Specify participation:
 - a. Partial for the Doctor being supervised as not every doctor will be supervised at all times.
 - b. Partial for the doctor doing the supervising as not all doctors will be supervising someone.
- 5.) If applicable, identify relationship attributes and justify why they belong to the relationship:
 - a. None again

Patient–Appointment:

- 1.) Provide the relationship name:
 - a. Patient were
- 2.) List participating entities:
 - a. Patient
 - b. Appointment
- 3.) Specify cardinality:
 - a. 1:N
- 4.) Specify participation:
 - a. Partial for patients as not every patient will have an appointment at all times.
A past patient might not have an appointment until they come in next or reach out to get something on the record.
 - b. Total participation from appointment as every appointment needs to have a patient.
- 5.) If applicable, identify relationship attributes and justify why they belong to the relationship:
 - a. None

Doctor–Appointment:

- 1.) Provide the relationship name:
 - a. Doctor to
- 2.) List participating entities:
 - a. Doctor
 - b. Appointment
- 3.) Specify cardinality:
 - a. 1:N
- 4.) Specify participation:
 - a. Total participation from appointment as there will always be at least one doctor for every appointment
 - b. Partial participation from doctor since every doctor can have many appointments.
- 5.) If applicable, identify relationship attributes and justify why they belong to the relationship:
 - a. None

Appointment–Medication:

- 1.) Provide the relationship name:
 - a. Medication from
- 2.) List participating entities:
 - a. Appointment
 - b. Medication
- 3.) Specify cardinality:
 - a. M:N
- 4.) Specify participation:
 - a. Partial from appointment as an appointment does not need to result in medication
 - b. Partial from medication as a medication can be prescribed during many appointments and different medications can be prescribed during a single appointment.
- 5.) If applicable, identify relationship attributes and justify why they belong to the relationship:
 - a. Dosage: Shown under the medication's description
 - b. Duration: Shown under the medication's description

Task 3 – Weak Entity Analysis:

The analyzation of the following entities is to determine if they should be weak or strong when modeled. In my attributes for task 1 I determined that they can be modeled as strong by assigning a globally unique key, however, not that I am looking at it I do not know if that is the proper strategy. In this task I will go through the list and determine the answer. Starting with Department since it's labeled first.

Department analysis:

- 1.) Is its key globally unique?
 - a. I modeled it with a unique key, but I do not believe it needs to have one since every department will be different per hospital the overlap from hospital to hospital will never be an issue through the association of the unique hospital that each department is assigned to.
- 2.) Is it existence-dependent?
 - a. Yes, the department is dependent to the hospital and as such it is existence-dependent. For every department must belong to one hospital.
- 3.) Does it require an identifying relationship?

- a. Yes, if it does not have a unique global key such as Department-Id then it would need some other key to determine uniqueness like the Hospital-Id and the Department Name.
- 4.) Final decision: Strong or Weak?
- a. This entity is weak and should be modeled as weak. If a user is searching for the department and they are already within the hospital, then it will be a locally unique name if they are searching for the department without a hospital globally then each department of the same name will be visible. With correct programming what hospital they are from will also be visible and as such the user will be presented with multiple options of where they need to go. Then they can choose the hospital and now they are to the unique department within a specific hospital. To find a department a user will already need to search for the dependent relationship that is tethered to department which is hospital.

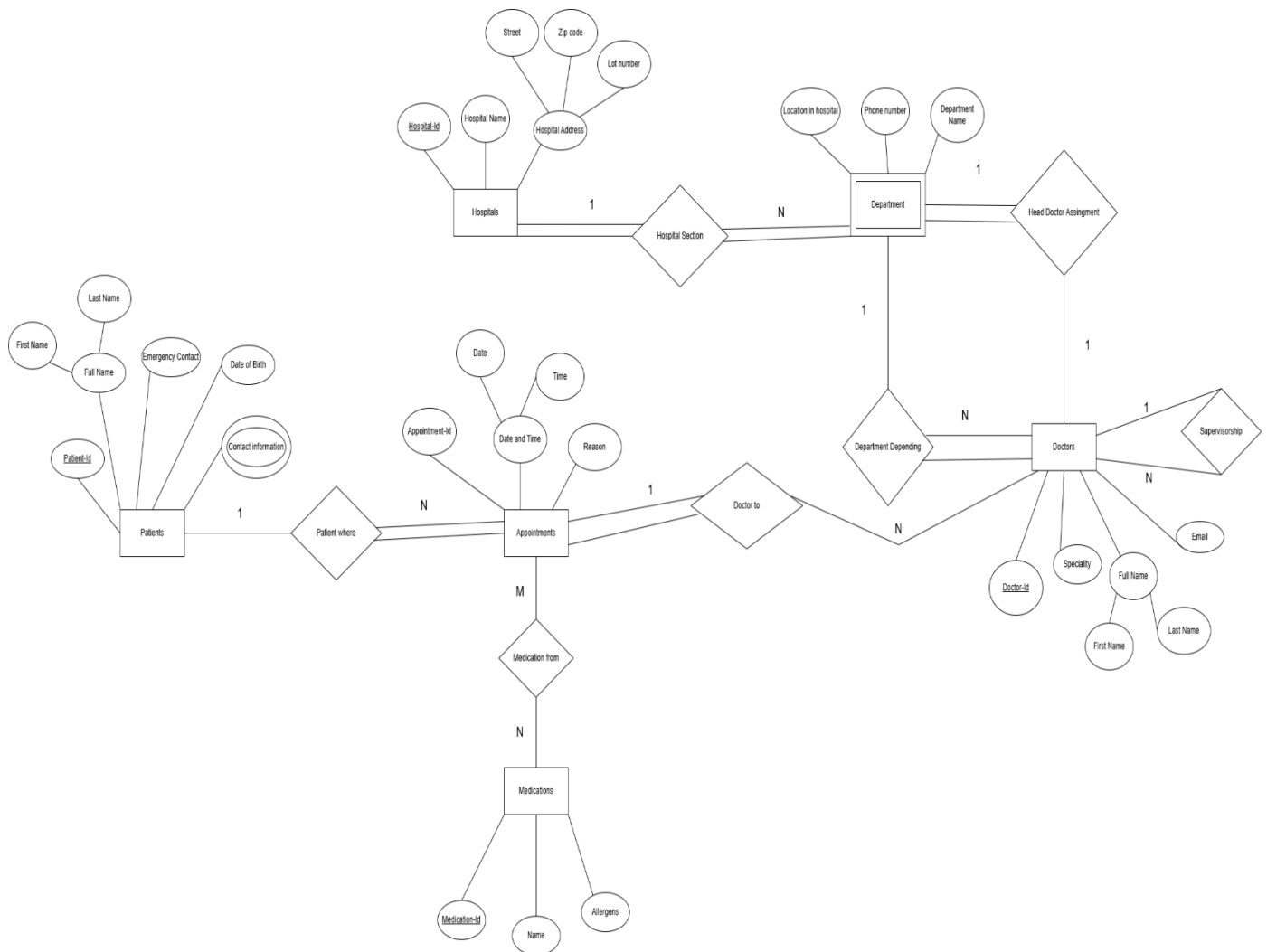
Appointment analysis:

- 1.) Is its key globally unique?
- a. In my task 1 I did make the appointment have a unique global attribute just as I did for the department. However, unlike department I believe it is still correct for this entity to have a globally defined unique attribute key.
- 2.) Is it existence-dependent?
- a. Yes, since an appointment cannot exist without one patient and one doctor exactly as stated in the assignment first page. If there is no patient or no doctor, then no appointment can exist even if both a patient and doctor exist it does not mean that the appointment must.
- 3.) Does it require an identifying relationship?
- a. It does not. Because the appointment has a unique global attribute key and it should have one then no other information from another source is needed to identify the appointment. So, while the appointment is existence-dependent it is not identity-dependent.
- 4.) Final decision: Strong or Weak?
- a. Appointment should be modeled as a strong entity and should be. Due to the fact that it is helpful for the appointment to have a unique global key and it does not need to have other information from other sources to be identified. Additionally, a patient and a doctor can have many appointments while for the department analysis each department is unique to a hospital so it makes sense to be weak the appointment must be strong since there can be many

instances of an appointment that are directly related to the entities that might interact with it.

Task 4 – Final ER Diagram:

For this portion of the assignment, I'm going to use draw.io to create the diagram. I uploaded a version of it here, but I will also upload it to GitHub as it is difficult to see everything in the diagram. However, it seems to be legible enough to make out what is being said. But if it's not there will be a larger png in GitHub.



Task 5 – Database Schema and Instance examples:

Producing the database schema and populating the different tables while staying true to the relationships means including primary and foreign keys. I will start with the Hospitals table below:

<u>Hospital-Id</u>	Hospital Name	Address
Hospital 1	Grand central	503 water street
Hospital 2	Beth beauty	490 gilded way
Hospital 3	Saint August	300 park place
Hospital 4	Peter the great	218 main street

Department table:

Good to note that two of the different hospitals have the same department but because they are under different hospitals they are still unique.

<u>Head Doctor-Id</u>	<u>Hospital-Id</u>	<u>Department Name</u>	Location in hospital	Phone number
Head Doctor 1	Hospital 2	Cardiology	West wing	333-289-1187
Head Doctor 2	Hospital 3	Surgery	East Wing	545-237-8809
Head Doctor 3	Hospital 1	Surgery	West wing	265-890-0379

Doctor table:

<u>Doctor-Id</u>	<u>Department Name</u>	<u>Hospital-Id</u>	Full Name	Specialty	Email	<u>Supervisor - Id</u>
Doctor 1	Cardiology	Hospital 2	Dr. Bimpilious Sniff	Heart adding	Sniff@Work.mail	Head Doctor 3
Doctor 2	Surgery	Hospital 3	Dr. Josh diamond	Nose Surgery	Diamond@Work.mail	None
Doctor 3	Surgery	Hospital 1	Dr. Pilfer Shout	Spinal Surgery	Shout@Work.mail	None
Doctor 4	Dermatology	Hospital 1	Dr. Amber Stone	Dermatology	Stone@Work.mail	Head Doctor 1

Patient table:

<u>Patient-Id</u>	Full name	Emergency contact	Date of birth	Contact information
Patient 1	Dralt Grimtolk	A dog he knows	07-23-800	Grimtolk@Gmail.com
Patient 2	Fish Masher	Fish Masher wife	08-4-1904	By Letter
Patient 3	John Appleseed	Mom	11-15-1980	By Bird
Patient 4	Bob (no last name)	Bob (no last name)	Unknown	Cannot be contacted

Appointment table:

<u>Appointment-Id</u>	<u>Patient-Id</u>	<u>Doctor-Id</u>	Date and time	Reason
Appointment 1	Patient 1	Doctor 1	12-20-2026, 08:00	Wants another heart
Appointment 3	Patient 2	Doctor 3	11-15-2026, 09:30	No more spine
Appointment 2	Patient 3	Doctor 2	02-06-2026, 11:15	Much larger node
Appointment 4	Patient 4	Doctor 4	09-04-2026, 23:45	Question about skin

Medication Table:

<u>Appointment-Id</u>	<u>Medication-Id</u>	Name	Allergens	Dosage	Duration
Appointment 1	Medicine 1	Almond medicine	Contains nuts	One big pill	10 years
Appointment 1	Medicine 2	Yummy medicine	Nothing is a sugar pill	Two big pills	20 years
Appointment 2	Medicine 3	Everything medicine	Contains everything ever	Small pill	1 minute